Show, Attend and Tell:
Re-implementing Soft Visual Attention for Image Captioning

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CS 4782 — Cornell University, Spring 2026

May 12, 2026

Introduction

Image captioning sits at the intersection of vision and language: a model must both *perceive* an image and *describe* it in fluent natural language. Pre-attention captioning approaches compressed the image into a single feature vector, forcing the decoder to remember every visual detail through a recurrent bottleneck. *Show, Attend and Tell* [1] (Xu et al., ICML 2015) broke this bottleneck by introducing **soft visual attention**: at every generated word, the decoder learns *where to look* in the image, producing both more accurate captions and per-word attention maps that are directly interpretable. This mechanism has since become foundational in modern vision-language systems (CLIP, BLIP, ViT-based VLMs). We re-implement the deterministic soft-attention variant on MS COCO 2014 to (1) reproduce the BLEU and METEOR numbers in Table 1 of the paper, and (2) verify the central qualitative claim — that attention maps reveal *where the model looks* when producing each word.

Chosen Result

We target the **MS COCO soft-attention row of Table 1** in Xu et al.: BLEU-1 = 70.7, BLEU-2 = 49.2, BLEU-3 = 34.4, BLEU-4 = 24.3, METEOR = 23.90. This is the most directly testable expression of the paper's central claim that soft attention is both an accuracy improvement *and* an interpretability tool.

Methodology

Our re-implementation matches the paper's encoder–decoder structure (Figure 1) with paper-faithful dimensions throughout. The **encoder** is a frozen VGG-19 [4], with features taken from `conv5_4` (the last conv layer before the final max-pool); a 224×224 input yields a 14×14×512 map, reshaped into $L = 196$ annotation vectors $a_i \in \mathbb{R}^{512}$. The **attention module** is a two-layer MLP that scores each a_i against the previous LSTM hidden state, softmaxed into weights $\alpha_{t,i}$; a learned scalar gate $\beta_t = \sigma(f_\beta(h_{t-1}))$ modulates the context vector $\hat{z}_t = \beta_t \sum_i \alpha_{t,i} a_i$ (Eq. 13). The **decoder** is a single-layer LSTM with hidden = embedding = 512; initial states (h_0, c_0) come from the mean annotation vector via two MLPs (Eq. 5–6), and the LSTM cell takes $(Ey_{t-1}, h_{t-1}, \hat{z}_t)$ as inputs (Eq. 1–2). The **deep output** layer predicts $p(y_t) \propto \text{softmax}(L_o(Ey_{t-1} + L_h h_t + L_z \hat{z}_t))$ over a 10 000-word vocabulary (Eq. 7). The **training loss** is cross-entropy plus doubly stochastic attention regularization with $\lambda = 1$ (Eq. 14). We train with Adam ($\text{lr} = 4 \times 10^{-4}$), batch size 128, dropout $p = 0.5$, teacher forcing, and early-stop on validation BLEU-4 (best at epoch 9). Evaluation uses beam search width 3, BLEU-1..4 without brevity penalty (paper convention), and **METEOR**.

Modifications from the original: PyTorch (vs. Theano), and the Karpathy splits [3] (the modern community standard for COCO captioning since 2015).

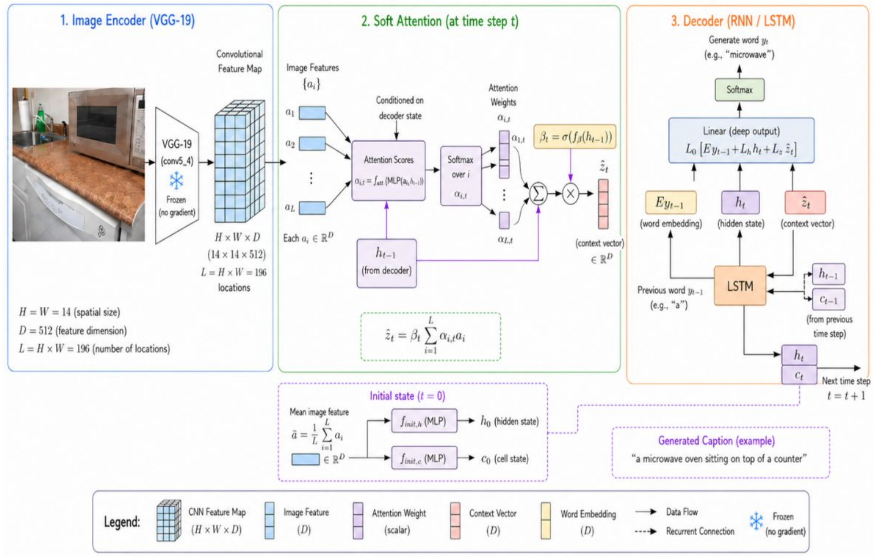

Figure 1: Soft-attention captioning architecture.

Results and Analysis

Quantitative. Table 1 compares our results to the paper. The headline metric BLEU-4 matches within **0.53 points**. More informative than the absolute gaps is the *pattern*: the gap **narrows monotonically** as n-gram order increases, from −7.11 at BLEU-1 to −0.53 at BLEU-4. This is unusual — higher-order BLEU is typically harder to match. The pattern instead suggests we match the paper's **fluency** (n-gram structure) but trail slightly on **word choice**. METEOR — which rewards synonym and stem matches — picks up the same word-choice signal as BLEU-1, hence the parallel ∼10% gap on those two metrics.

Metric	Paper	Ours	Δ	Relative gap
BLEU-1	70.70	63.59	−7.11	10.1%
BLEU-2	49.20	45.79	−3.41	6.9%
BLEU-3	34.40	32.70	−1.70	4.9%
BLEU-4	24.30	**23.77**	**−0.53**	**2.2%**
METEOR	23.90	21.55	−2.35	9.8%

Table 1: Karpathy test-split results vs. Xu et al. Table 1 (soft-attention MS COCO row).

Qualitative captions. Figure 2 shows a representative success and a canonical failure. Successful captions are syntactically fluent and capture the major objects in the scene, often reproducing reference captions almost verbatim. Failures cluster around two well-documented limitations of cross-entropy-trained captioners: object confusion in cluttered or unfamiliar scenes (most strikingly, a single llama captioned as "a herd of sheep standing on top of a grass covered field").

Attention behavior. Beyond caption-level metrics, we examined the paper's central interpretability claim by visualizing the per-word attention distribution on a separate test example (Figure 3). Attention concentrates sharply on the relevant image region for content words ("hydrant," "yellow") and disperses for function words ("a," "on," "the"). This pattern reproduces the qualitative behavior reported by Xu et al. and confirms that soft attention produces visualizations that are directly interpretable.

Reflections

Evaluation tooling matters more than expected. Our METEOR results were ∼41.2 — nearly 2× the paper's 23.90 — because NLTK's default `meteor_score` returns averages on a different scale than the Java `meteor-1.5.jar` used in the paper. Recomputing with the Java reference produced 21.55, which is consistent

(a) Generated: *a baseball player swinging a bat at a ball*
Reference: *a baseball player up to plate about to hit the ball*

(b) Generated: *a herd of sheep standing on top of a grass covered field*
Reference: *a llama looks at the camera from a fenced in, grassy area*

Figure 2: Representative generated captions on the Karpathy test split: (a) a successful case where the model produces a fluent, accurate description; (b) a failure case where the model is confidently wrong.

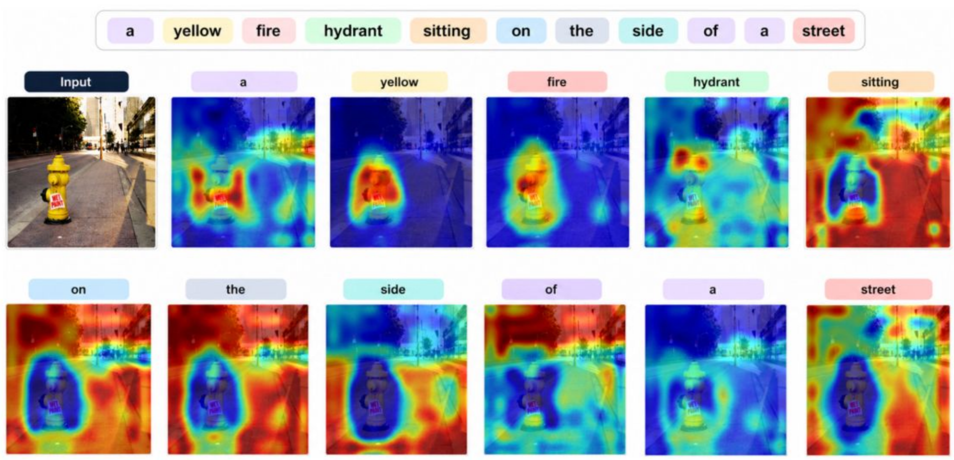

Figure 3: Per-word attention visualization for one test image.

with the rest of our metrics. We added a paper-comparable rescoring script (`code/recompute_meteor.py`) to the repository to prevent this silent failure mode from recurring.

Qualitative and quantitative evaluation are complementary. BLEU/METEOR give compact comparison numbers, but attention maps reveal what the model is *actually doing*. In our case, the per-word attention visualizations provided direct evidence that the model attends to relevant image regions during generation, a diagnostic insight quantitative metrics alone could not have surfaced.

Future directions. With more time, we would: (1) replace VGG-19 with a ViT or CLIP backbone to test whether the residual BLEU-1/METEOR gap is encoder-bound; (2) replace the fixed 14×14 grid with object-region attention [5] so the model attends to detected entities rather than arbitrary patches; (3) fine-tune with CIDEr-optimized self-critical training [6] to reduce exposure bias; and (4) directly compare against the hard-attention variant to study whether its interpretability properties carry over.

References

[1] K. Xu, J. L. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, Y. Bengio. *Show, Attend and Tell: Neural Image Caption Generation with Visual Attention.* ICML, 2015.

[2] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C. L. Zitnick. *Microsoft COCO: Common Objects in Context.* ECCV, 2014.

[3] A. Karpathy and L. Fei-Fei. *Deep Visual-Semantic Alignments for Generating Image Descriptions.* CVPR, 2015.

[4] K. Simonyan and A. Zisserman. *Very Deep Convolutional Networks for Large-Scale Image Recognition.* arXiv:1409.1556, 2014.

[5] P. Anderson, X. He, C. Buehler, D. Teney, M. Johnson, S. Gould, L. Zhang. *Bottom-Up and Top-Down Attention for Image Captioning and Visual Question Answering.* CVPR, 2018.

[6] S. J. Rennie, E. Marcheret, Y. Mroueh, J. Ross, V. Goel. *Self-Critical Sequence Training for Image Captioning.* CVPR, 2017.

[7] A. Paszke, S. Gross, F. Massa, et al. *PyTorch: An Imperative Style, High-Performance Deep Learning Library.* NeurIPS, 2019.